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To the Chief Editor,
Nature Catalysis
Editorial Office
Springer Nature Publishing Group

Subject: Submission of Original Research Article for publication in Nature Catalysis

Dear Editor,

We are submitting our original research manuscript entitled “**A physics-informed matrix engine bypasses scaling relations in alkyne-poisoned catalytic interfaces**” for consideration as a Research Article in *Nature Catalysis*.

A central goal in modern heterogeneous catalysis is designing interfaces that can handle trace impurities without dropping in performance. In industrial olefin purification streams, terminal alkyne impurities selectively bind to and permanently poison active metal sites by forming irreversible σ -acetylides. Bypassing this problem using traditional computational screening has been fundamentally blocked by linear scaling relations. These relations enforce a strict thermodynamic link between benign intermediate binding and destructive deactivation pathways. Furthermore, standard static metrics like the surface d -band center cannot distinguish between these two modes of interaction.

In this work, we address this long-standing bottleneck by introducing a fast, analytical quantum screening model based on a modified three-center Hückel-Coulson Hamiltonian matrix. By operating at sub-millisecond speeds per catalyst-reactant pair, our physics-informed engine calculates dynamic electronic properties: π -backbonding susceptibility (ΔE_π) and terminal carbon Coulson charge redistribution (Δq_{C1}).

By screening 300,000 distinct configurations, our model maps a clear electronic phase boundary that separates safe, reversible coordination from irreversible proton deactivation. Using this phase map, we identify two distinct structural classes—coinage-metal single-atom alloys (such as $Pd_1/Ag(111)$) and spatially confined catalysts (such as Fe_1-N_4)—that completely break traditional scaling trends via local charge inversion. Finally, we validate these discoveries using microkinetic Kinetic Monte Carlo (kMC) simulations. These tests show that our engineered interfaces maintain over 98.5% active-site availability under competitive industrial operating environments.

Given that *Nature Catalysis* prioritizes fundamental mechanistic breakthroughs that shift current design paradigms, we believe this work will be of great interest to your readers. By replacing uninterpretable data-driven correlations with explicit chemical intuition, this model provides an adaptable, computationally scalable framework for designing durable catalytic interfaces across many multi-center chemical environments.

This manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the submission. Thank you for your time and consideration.

Sincerely,

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